

Name: Dylan M, Arnav, Marcus S

Date: 3/11/2026

Class: Engineering Design

MyDesign Portfolio

Element B

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <Wood Block used in Woodshop Room>



Figure 1 - Wood Block Cover A



Figure 2 - Wood Block Cover B

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Figure 3 - Mr D inside the woodshop room using wooden block in port

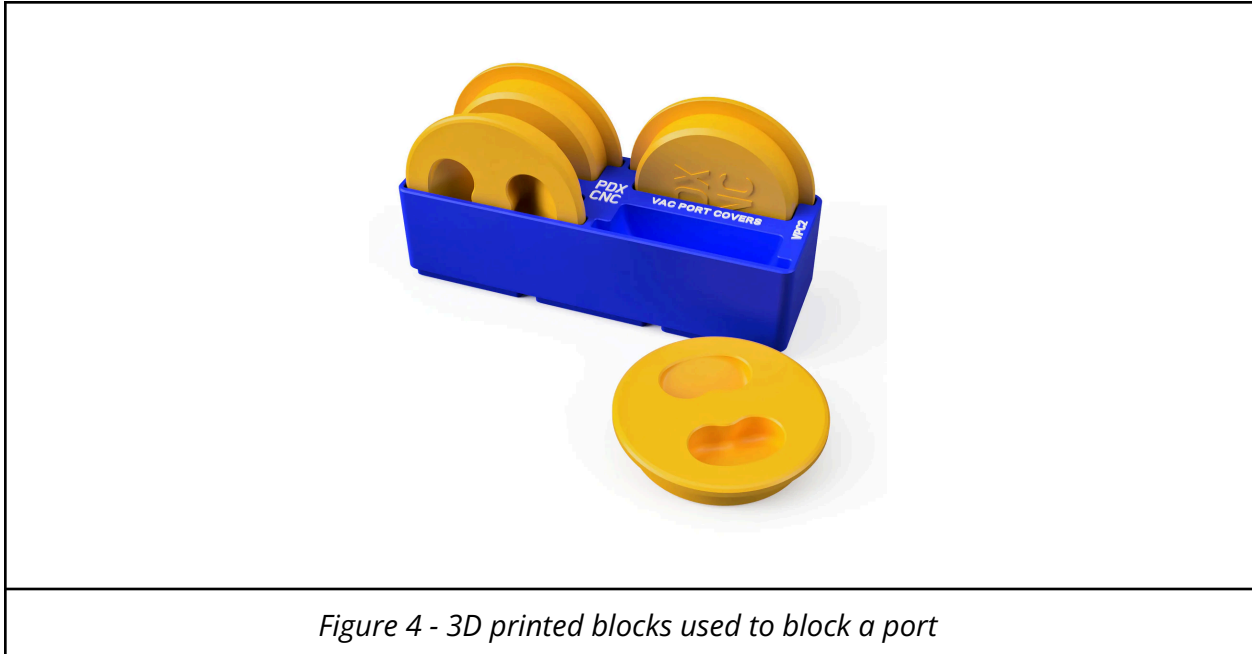
This design uses a wooden block (shown in figure 1) in the Walpole High School's woodshop room. This design is efficient, but does not not seem or look safe, and not appealing. This can be redesigned to look better, and create a fitted cover, rather than a blocky piece of wood to control the flow of air.

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Prior Design Attempt #2 - <3D Printed Port Covers>



Explain: These port covers are a 3D printed cover that can be used to block the air flow through a port. This design uses 3D printed material, so it is not as strong as a bigger piece of wood, but still is durable, good looking, and an efficient way to design a cover for air flow through a port. For a pack of 4, It costs \$15, which is not a bad price as it is fairly affordable.

This can help us get an idea of what we would like ours to appear and function. It has latches that you can use for your fingers to open and close these ports easily, making it very friendly to use

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Prior Design Attempt #3 - <Wall connector closing port #3>



Figure 5 - Closable port used to block air

This design uses a wall mount and a closing extension to open and close the port to control the flow of air when the machine is running. When running the dust collector and the sander is not in use, the person can shut this port so they can get more efficient flow throughout the system, but also have a clean and breathable room.

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Prior Design Attempt #4 - <Wood block #1>



Figure 6 - A wooden block used to fit a smaller block to stop flow

This design uses a block that has holes in it. The top will be connected to the dust collector system and the holes on the side can be covered up by the wooden blocks or be connected to a vacuum hose. This design was unsuccessful because the wooden blocks made for the hole didn't properly fit in the holes. As a result, the blocks didn't fully block outside air flow, which is what our stakeholders need. Additionally, the design was too heavy for our stakeholder's requirements.

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EXTRA IDEAS:

[Custom Vacuum Port Cap with wire tether](#)

[WoodRiver Dust Connection Port Cap 4" OD for Dust Collection Systems | Woodcraft](#)

Works Cited:

Figure 1- Sheldon, Marcus. Taken 3/8/26.

Figure 2 - Sheldon, Marcus. Taken 3/8/26.

Figure 3 - Sheldon, Marcus. Taken 3/8/26.

Figure 4 - [CNC Router Vacuum Port Cover](#)

Figure 5 - [Small Shop Dust Collection \(What I Chose and Why\) | Saws on Skates®](#)

Figure 6 [Dust collection cover plates for 4-way outlet](#)